

Final Assessment - Practical – Development/Programming Assessment

(Cathleen Dayao)

Marks Break Down:

Section A: Question 1: 15 Marks

Section A: Question 2: 15 Marks

Section B: Question 1: 15 Marks

Section B: Question 2: 15 Marks

GitHub Profile – End to End - Assignments : 40 Marks

Course: Fullstack AI – Batch – Cathleen Dayao

**You have to program/Develop : 2 question dataset from Section A and
2 question from section B**

Theme of Paper: Philippines

References:

<https://psa.gov.ph>
<https://www.tourism.gov.ph>
<https://www.ibpap.org>
<https://www.pna.gov.ph>
<https://www.philstar.com>

1. **Population Size**

The Philippines has a population of over **115 million people**, making it one of the largest consumer and labor markets in Southeast Asia.

2. **Strong Economic Growth**

The country is among the major emerging economies in Asia, supported by services, manufacturing, remittances, and digital industries.

3. **Large English-Speaking Workforce**

The Philippines is recognized globally for its high English proficiency, which helps attract international technology and outsourcing companies.

4. **Growing Software Development Sector**

The software industry continues to expand through application development, cloud services, cybersecurity, AI solutions, and enterprise software projects.

5. **Abundant Technical Talent**

Universities produce thousands of graduates annually in IT, computer science, engineering, and related fields, creating a strong talent pipeline for software companies.

6. **Global Outsourcing Leader**

The Philippines is one of the world's leading destinations for Business Process Outsourcing (BPO) and IT-enabled services.

7. **IT-BPM Revenue Growth**

The Philippine IT-BPM (Information Technology and Business Process Management) industry generated approximately **US\$38 billion in revenue in 2024**, reflecting strong international demand.

8. **Major Employment Generator**

The IT-BPM sector employed around **1.82 million people in 2024**, making it one of the country's largest private-sector employers.

9. **Key Offshore Services Provided**

Offshore services include customer support, software development, IT helpdesk services, finance and accounting, healthcare support, data analytics, and digital marketing.

10. **Future Growth Driven by AI and Digital Skills**

Industry groups are investing in upskilling workers in artificial intelligence, automation, and digital technologies to maintain the Philippines' competitiveness as a global outsourcing hub.

=====Dataset:

Dataset 1. [Philippine - Data Analyst Remote Jobs]

<https://github.com/ShahzadSarwar10/FULLSTACK-AI-1-TO-1-SESSION-SAT-SUN-Philippines/tree/main/FinalAssessment/Data%20Analyst%20Remote%20Jobs>

Dataset 2.[Philippine - Data Salaries]

<https://github.com/ShahzadSarwar10/FULLSTACK-AI-1-TO-1-SESSION-SAT-SUN-Philippines/tree/main/FinalAssessment/Data%20Salaries>

Dataset 3:[Philippine - EV Charging Stations]

<https://github.com/ShahzadSarwar10/FULLSTACK-AI-1-TO-1-SESSION-SAT-SUN-Philippines/tree/main/FinalAssessment/EV%20Charging%20Stations>

Dataset 4:[Philippine - Earthquakes 1900-2026]

<https://github.com/ShahzadSarwar10/FULLSTACK-AI-1-TO-1-SESSION-SAT-SUN-Philippines/tree/main/FinalAssessment/Earthquakes%201900-2026>

Dataset 5:[Philippine - Real Estate & Airbnb Data]

<https://github.com/ShahzadSarwar10/FULLSTACK-AI-1-TO-1-SESSION-SAT-SUN-Philippines/tree/main/FinalAssessment/Real%20Estate%20%26%20Airbnb%20Data>

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Section A: Machine Learning - SK LEARN [Marks 15+15]

Need to apply – Numpy, Pandas, Seaborn

Apply: Classification, Regression, Clustering, Ensemble Learning

Question/Task - 1:

Select one of the above dataset and apply.

Question/Task 2:

Next, Select one of the above different dataset and apply.

Steps - TO Do: [Neat, Clean, well-documented code]

Loading in DataFrame, Perform “Descriptive Statistics” Max- Operations, Input data analysis based on seaborn, Performing core operation of SK Learn, Perform metric analysis

[10 Marks for exceptional creative deep data analytical skills demonstration. For standard implementation – these marks will NOT be given.]

Need demonstration of all skills, learn in this course. End to end. For standard implementation, no marks , will be given. AI generated code or copy pasted code, will get only zero marks.

Section B: Deep Learning [Marks 15 + 15]

Process, All 2 dataset

Apply end to end, Apply – Numpy, Pandas, SeaBorn

Apply Time SERIES – RNN , LSTM and GRU

Steps - TO Do:[Neat, Clean, well-documented code]

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Design Model with related layers, Performing core operation of Tensflow-Keras based deep learning, Perform Model metric analysis

[10 Marks for exceptional creative deep data analytical skills demonstration. For standard implementation – these marks will NOT be given.]

Question/Task - 3:

Select one of the above dataset and apply.

Question/Task 4:

Next, Select one of the above different dataset and apply.

Need demonstration of all skills, learn in this course. End to end. For standard implementation, no marks , will be given. AI generated code or copy pasted code, will get only zero marks.